Power BI Data Transformation and Data Modeling

Description

This Power BI project focuses on transforming, cleaning, and modeling sales data collected from three datasets: List of Orders, Order Details, and Sales Target. The project demonstrates how Power Query and Data Modeling techniques can be used to prepare raw data for reporting and business analysis. Various transformation methods such as data type conversion, text formatting, column creation, merging, filtering, grouping, and relationship building were applied to create a structured and analysis-ready dataset.

Project Objectives

The main objectives of this project are:

- To import and transform multiple datasets in Power BI.
- To correct data types and improve data consistency.
- To create new calculated columns for business analysis.
- To merge and organize data from different sources.
- To identify and validate data quality issues.
- To perform grouping and aggregation operations.
- To establish relationships between tables using data modeling.
- To prepare a clean dataset for dashboard creation and reporting.

Dataset Description

The project uses the following datasets:

- Order ID
- Order Date
- Customer Name
- City
- State
- Category
- Sub-Category

- Amount
- Profit
- Quantity
- Target

These datasets were transformed and connected to support sales and profit analysis.

Data Cleaning and Transformation Tasks

Data Import

All source files were imported into Power BI and loaded into Power Query Editor for transformation and preparation.

Data Type Conversion

Appropriate data types were assigned to important fields.

Tasks Performed:

- Converted Order Date into Date format.
- Changed Amount and Target columns into Fixed Decimal Number format.
- Ensured data accuracy for calculations and reporting.

Text Standardization

Text values were formatted to maintain consistency.

Tasks Performed:

- Applied Capitalize Each Word to Customer Name.
- Improved readability and standardized naming conventions.

Creating New Columns

Additional columns were created to support analysis.

Tasks Performed:

- Merged City and State into a new Location column.
- Calculated Profit Margin using Profit and Amount values.
- Converted Profit Margin into Percentage format.

- Created Profit Status column to classify records as Profit, Loss, or Break-Even.

Data Integration

Multiple datasets were combined to create a unified data model.

Tasks Performed:

- Merged List of Orders and Order Details using Order ID.
- Expanded required columns from the merged table.
- Verified successful integration of records.

Data Validation

Data quality checks were performed to ensure reliability.

Tasks Performed:

- Checked for missing values.
- Verified column quality.
- Identified duplicate records.
- Confirmed dataset consistency after transformations.

Sorting and Filtering

Data organization techniques were applied.

Tasks Performed:

- Sorted records by Order Date in descending order.
- Filtered records based on selected state values for regional analysis.

Grouping and Aggregation

Aggregation functions were used to generate business insights.

Tasks Performed:

- Counted total orders.
- Calculated average profit by category.
- Summed sales amount by sub-category.
- Aggregated monthly sales targets.

Data Modeling

Relationship Creation

Relationships were established between tables using the Manage Relationships feature.

Tasks Performed:

- Connected tables through Order ID and other relevant fields.
- Built a structured data model for reporting.
- Improved analysis efficiency and accuracy.

Tools and Features Used

The following Power BI features were utilized:

- Power Query Editor
- Data Type Transformation
- Merge Queries
- Custom Columns
- Conditional Columns
- Sorting and Filtering
- Group By Operations
- Data Validation
- Relationship Management
- Data Modeling

Descriptive Analysis (What Happened?)

The Power BI project analyzed 500 sales records from multiple datasets. Data was cleaned, transformed, and merged to create a structured dataset. Sales, profit, and target performance were analyzed to provide insights into overall business performance.

Diagnostic Analysis (Why Did It Happen?)

Profitability varied across categories and sub-categories. Furniture generated the highest average profit, while Bookcases and Printers contributed the most sales. Regional differences and customer demand influenced sales and profit performance.

Predictive Analysis (What Is Likely to Happen?)

High-performing categories and products are expected to continue generating strong sales and profits. Regions with consistent growth are likely to maintain their performance, and sales targets may increase as business expands.

Prescriptive Analysis (What Should Be Done?)

Focus on promoting profitable products, maintain inventory for top-selling items, and review loss-making transactions. Regularly monitor sales against targets and use dashboard insights to support better business decisions and growth.

Conclusion

This project successfully transformed raw sales data into a clean and structured dataset suitable for analysis. By applying Power Query transformations and building relationships between tables, the data became ready for creating interactive dashboards, performance reports, and business insights within Power BI.